

Supplementary File 2

PubMed/MEDLINE search strategy

Search executed: 15 May 2026 (PubMed/MEDLINE via NCBI E-utilities).
Date range: 1 January 2021 to 15 May 2026.
Records returned: 245.
Filters applied: None at search level (filters applied at title/abstract and full-text screening).

Conceptual blocks

Four conceptual blocks combined with the AND operator. Within each block, terms were combined with OR. Block construction:

Block	Terms
Foundation models	"foundation model", "vision-language model", "BiomedCLIP", "MedCLIP", "RadFM", "RadImageNet", "Merlin", "MedSAM", "Segment Anything", "self-supervised learning", "masked image modeling", "contrastive learning", "pretrained model", "transfer learning"
Radiomics	"radiomics", "radiomic", "PyRadiomics", "handcrafted features", "GLCM", "gray-level co-occurrence", "GLRLM", "texture features", "IBSI", "MIRP", "LIFEx", "MaZda"
Imaging modality	"computed tomography", "CT scan", "MRI", "magnetic resonance", "PET", "ultrasound", "X-ray", "radiograph", "mammography", "histopathology", "digital pathology", "medical imaging"
Comparison	"comparison", "compared", "versus", "vs", "benchmark", "benchmarking", "head-to-head", "performance comparison"

Full Boolean expression

("foundation model"[TIAB] OR "vision-language model"[TIAB] OR "BiomedCLIP"[TIAB] OR "MedCLIP"[TIAB] OR "RadFM"[TIAB] OR "RadImageNet"[TIAB] OR "Merlin"[TIAB] OR "MedSAM"[TIAB] OR "Segment Anything"[TIAB] OR "self-supervised learning"[TIAB] OR "masked image modeling"[TIAB] OR "contrastive learning"[TIAB] OR "pretrained model"[TIAB] OR "transfer learning"[TIAB]) AND ("radiomics"[TIAB] OR "radiomic"[TIAB] OR "PyRadiomics"[TIAB] OR "handcrafted features"[TIAB] OR "GLCM"[TIAB] OR "gray-level co-occurrence"[TIAB] OR "GLRLM"[TIAB] OR "texture features"[TIAB] OR "IBSI"[TIAB] OR "MIRP"[TIAB] OR "LIFEx"[TIAB] OR "MaZda"[TIAB]) AND ("computed tomography"[TIAB] OR "CT scan"[TIAB] OR "MRI"[TIAB] OR "magnetic resonance"[TIAB] OR "PET"[TIAB] OR "ultrasound"[TIAB] OR "X-ray"[TIAB] OR "radiograph"[TIAB] OR "mammography"[TIAB] OR "histopathology"[TIAB] OR "digital pathology"[TIAB] OR "medical imaging"[TIAB]) AND ("comparison"[TIAB] OR "compared"[TIAB] OR "versus"[TIAB] OR "vs"[TIAB] OR "benchmark"[TIAB] OR "benchmarking"[TIAB] OR "head-to-head"[TIAB] OR "performance comparison"[TIAB]) AND ("2021/01/01"[PDAT] : "2026/05/15"[PDAT])

Note on databases: Per PROSPERO record CRD420261393443, the review was scoped to PubMed/MEDLINE only. Embase, Web of Science and grey-literature searches were not run; this limitation is acknowledged in §5.4 of the manuscript.